

DARBOUX, MOSER AND WEINSTEIN THEOREMS FOR PREQUANTUM SYSTEMS AND APPLICATIONS TO GEOMETRIC QUANTIZATION

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ABSTRACT. We establish analogs of the Darboux, Moser and Weinstein theorems for prequantum systems. We show that two prequantum systems on a manifold with vanishing first cohomology, with symplectic forms defining the same cohomology class and homotopic to each other within that class, differ only by a symplectomorphism and a gauge transformation. As an application, we show that the Bohr-Sommerfeld quantization of a prequantum system on a manifold with trivial first cohomology is independent of the choice of the connection.

1. INTRODUCTION

The Darboux theorem establishes that there are no local invariants in symplectic geometry. Namely, let M be a smooth manifold and let ω, ω' be two symplectic forms on M , then:

Theorem 1 (Darboux [1, 2]). *For every point $p \in M$ there exists a neighbourhood U of p and an embedding $\Phi : U \rightarrow M$ isotopic to the inclusion and fixing p such that*

$$\Phi^*\omega' = \omega|_U.$$

The global invariants of a symplectic manifold include the cohomology class defined by its symplectic form. This gives a complete classification of symplectic 2-manifolds. In general, there is the following theorem due to Moser.

Theorem 2 (Moser [6]). *Let M be a compact manifold endowed with two symplectic forms ω and ω' . Assume that $[\omega] = [\omega']$, and that there exists a path ω_t of symplectic forms such that $[\omega_t] = [\omega_0]$ for all t , and with $\omega_0 = \omega$ and $\omega_1 = \omega'$. Then, there exists a diffeomorphism isotopic to the identity $\Phi : M \rightarrow M$ such that $\Phi^*\omega' = \omega$.*

This proof based on Moser's method can be adapted when symmetries are present. Under the conditions above when a compact group acts on M preserving ω and ω' , and where the path ω_t also consists of invariant forms, we have the following result.

Theorem 3 (Weinstein [7]). *There exists an equivariant diffeomorphism $\Phi : M \rightarrow M$ such that $\Phi^*\omega' = \omega$.*

Recall that if (M, ω) is a symplectic manifold, $\pi : L \rightarrow M$ is a (complex) line bundle, ∇ is a connection on L , and $\text{curv}(\nabla) = \omega$, the quadruple (M, ω, L, ∇) is a *prequantum system*.

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The main purpose of this article is to generalize these three theorems to prequantum systems.

Theorem 4. *Suppose (M, ω, L, ∇) and (M, ω', L, ∇') are prequantum systems. Then for every point $p \in M$ there exists a neighbourhood U of p , an embedding $\Phi : U \rightarrow M$ isotopic to the inclusion and fixing p , and a function $\phi \in C^\infty(U, \mathbb{C})$ such that¹*

$$\begin{aligned}\omega|_U &= \Phi^* \omega' & (\Phi^* L) &= L|_U \\ \nabla s &= (\Phi^* \nabla') s + d\phi \otimes s, \text{ for any } s \in \Gamma(L|_U).\end{aligned}$$

If L is a hermitian line bundle and ∇ is a hermitian connection, we may take $\phi \in C^\infty(M, i\mathbb{R})$.

For compact manifolds, we have

Theorem 5. *Suppose (M, ω, L, ∇) and (M, ω', L, ∇') are prequantum systems with M a compact manifold. Assume that $[\omega] = [\omega']$, and that there exists a path ω_t of symplectic forms with $[\omega_t] = [\omega_0]$ for all t , such that $\omega_0 = \omega$ and $\omega_1 = \omega'$. Assume also that $H^1(M, \mathbb{R}) = 0$.*

Then there exists a diffeomorphism $\Phi : M \rightarrow M$ isotopic to the identity and a function $\phi \in C^\infty(M, \mathbb{C})$ such that

$$\begin{aligned}\omega &= \Phi^* \omega' & \Phi^* L &= L \\ \nabla s &= (\Phi^* \nabla') s + d\phi \otimes s, \text{ for any } s \in \Gamma(L).\end{aligned}$$

If L is a hermitian line bundle and ∇ is a hermitian connection, we may take $\phi \in C^\infty(M, i\mathbb{R})$.

When there are additional symmetries the following result holds.

Theorem 6. *Suppose G is a compact Lie group, and suppose M is a compact G -space. Suppose that (M, ω, L, ∇) and (M, ω', L, ∇') are G -invariant prequantum systems. Assume that $[\omega] = [\omega']$, and that there exists a path ω_t of G -invariant symplectic forms with $[\omega_t] = [\omega_0]$ for all t , such that $\omega_0 = \omega$ and $\omega_1 = \omega'$. Assume also that $H^1(M, \mathbb{R}) = 0$.*

Then there exists a G -equivariant diffeomorphism $\Phi : M \rightarrow M$ isotopic to the identity and a G -invariant function $\phi \in C^\infty(M, \mathbb{C})$ such that

$$\begin{aligned}\omega &= \Phi^* \omega' & \Phi^* L &= L \\ \nabla s &= (\Phi^* \nabla') s + d\phi \otimes s, \text{ for any } s \in \Gamma(L).\end{aligned}$$

If L is a hermitian line bundle and ∇ is a hermitian connection, we may take $\phi \in C^\infty(M, i\mathbb{R})$.

Remark 1.1. The function ϕ is called a *gauge transformation*. We have thus proven that two prequantum systems, with associated symplectic forms lying in the same cohomology class and homotopic to each other, differ only by a symplectomorphism and a gauge transformation. Observe that it is not possible to dispense with the gauge transformation: Suppose ∇ is the trivial connection on the trivialized bundle $L = M \times \mathbb{C}$, and ∇' is some other connection on L . Then for any section $s \in \Gamma(M, L)$, we have $\nabla' s = \nabla s + d\phi \otimes s$, where ϕ is somewhere non-constant. And no symplectomorphism will transform ϕ into an everywhere constant function (since diffeomorphisms take constant functions to constant functions).

As an application of this theorem, we prove that for manifolds with trivial first cohomology, the Bohr-Sommerfeld quantization of a prequantum system (M, ω, L, ∇) does not depend on the choice of the connection ∇ ; in other words if (M, ω, L, ∇') is another prequantum system associated with the same manifold, line bundle, and symplectic form, its

¹For the definition of the pullback of a connection under a smooth map, see e.g. [5], p. 292, Lemma 3.

Bohr-Sommerfeld quantization coincides with that of (M, ω, L, ∇) . If $H^1(M) \neq 0$ a similar result holds if the holonomy representations of ∇ and ∇' coincide.

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2. PROOF OF THE MAIN THEOREMS

We first prove Theorem 4.

Proof. By the Darboux theorem, we know that there exists a neighborhood U of p , which we may take to be diffeomorphic to an open ball, and an embedding $\phi : U \rightarrow M$ isotopic to the inclusion preserving p such that

$$\Phi^* \omega' = \omega|_U.$$

Since Φ is isotopic to the inclusion we also have

$$\Phi^* L = L|_U$$

It remains to compare the two connections and consider the difference

$$\nabla s - (\Phi^* \nabla') s \text{ for any } s \in \Gamma(L|_U).$$

Since the space of connections on a line bundle on U is an affine space modeled on $\Omega^1(U, \mathbb{C})$, we know that

$$\nabla s - (\Phi^* \nabla') s = \xi \otimes s$$

for some $\xi \in \Omega^1(U, \mathbb{C})$.

Since $\text{curv } \nabla = \text{curv } \Phi^*(\nabla')$, it follows that $d\xi = 0$. Since $H^1(U, \mathbb{R}) = 0$, $\xi = d\phi$ for some $\phi \in C^\infty(U, \mathbb{C})$. If the line bundle L is hermitian and the connection is unitary, we may repeat this argument to show that we may take $\phi \in C^\infty(U, i\mathbb{R})$. $\square$

We next prove Theorem 5.

Proof. The existence of the diffeomorphism is guaranteed by Moser's theorem. We again have $\Phi^* L = L$, and again there exists $\xi \in \Omega^1(M, \mathbb{C})$ so that

$$\nabla s - (\Phi^* \nabla') s = \xi \otimes s \text{ for any } s \in \Gamma(L).$$

Again $d\xi = 0$, so since $H^1(M, \mathbb{R}) = 0$, $\xi = d\phi$ for some ϕ , as needed. If the line bundle L is hermitian and the connection is unitary, we may repeat the argument to show that we may take $\phi \in C^\infty(M, i\mathbb{R})$. $\square$

We now proceed to the proof of Theorem 6.

Proof. The existence of a G -equivariant diffeomorphism Φ is given by Weinstein's theorem.

Since $H^1(M, \mathbb{R}) = 0$, again there exists a function ϕ such that

$$(2.1) \quad \nabla s - (\Phi^* \nabla') s = d\phi \otimes s \text{ for any } s \in \Gamma(L).$$

Let $\bar{\phi}$ be the average of ϕ over G using the Haar measure μ , i.e.,

$$\bar{\phi} := \int_G \alpha_g^* \phi \, d\mu(g),$$

where we have denoted by α_g the diffeomorphism giving the action of $g \in G$ on M .

Then since ∇ and ∇' are G -invariant, we obtain

$$\nabla s - (\Phi^* \nabla') s = d\bar{\phi} \otimes s$$

by averaging both sides of equation (2.1).

Again, if the line bundle L is hermitian and the connection is unitary, we may repeat this argument to show that we may take $\phi \in \mathcal{C}^\infty(M, i\mathbb{R})^G$. $\square$

Remark 2.2. (The case where $H^1(M, \mathbb{R}) \neq 0$) Suppose we are given two connections ∇ and ∇' on L with the same curvature. Then for any section $s \in \Gamma(M, L)$, we have $\nabla s = \nabla' s + \alpha \otimes s$ for some $\alpha \in \Omega^1(M, \mathbb{C})$ with $d\alpha = 0$.

If $H^1(M, \mathbb{C}) \neq 0$, we cannot conclude that $\alpha = d\phi$ for some function ϕ . However, if ∇ and ∇' also have the same holonomy representation, it follows that the holonomy representation of α , thought as a connection on the trivialized bundle $L \otimes L^* = \mathbb{C}$, is trivial. That means that for every oriented curve C in M , $\int_C \alpha = 2\pi n$, where $n \in \mathbb{Z}$. This does not imply that α is exact; but there exists a map $t : M \rightarrow S^1$ with $\alpha = t^*(d\theta)$, where $d\theta$ is the translation invariant one form on S^1 .²

3. APPLICATION TO QUANTIZATION IN A REAL POLARIZATION

Let (M, ω, L, ∇) be a prequantum system. Recall that a Lagrangian submanifold Λ is *integral* if $(L, \nabla)|_\Lambda$ is trivial as a bundle with connection.

Corollary 1. *Let M be a compact manifold with $H^1(M, \mathbb{R}) = 0$ and let (M, ω, L, ∇) and (M, ω, L, ∇') be two prequantum systems. A Lagrangian submanifold $\Lambda \subset M$ is integral for (M, ω, L, ∇) if and only if it is integral for (M, ω, L, ∇') .*

In particular, suppose we are given a real polarization of M , that is, a foliation by Lagrangian submanifolds. Consider from now on prequantum systems (M, ω, L, ∇) where L is hermitian and the connection ∇ is unitary. The Bohr-Sommerfeld quantization of (M, ω, L, ∇) associated to this polarization is given by the vector space generated by the integral leaves of the foliation, or equivalently by the covariant constant sections of (L, ∇) on those leaves.

In [3] Guillemin and Sternberg consider the case where the foliation is given by a map $\pi : M \rightarrow B$, where B is a simply connected subset of $\mathbb{R}^n$ and the map $F : M \rightarrow B$ is the moment map for an integrable system away from the singularities of the foliation. The Bohr-Sommerfeld quantization is then determined by the integral points of the image of the moment map F .

Then Corollary 1 implies

Corollary 2. *Let (M, ω) be a symplectic manifold with $H^1(M, \mathbb{R}) = 0$. Let L be a complex line bundle on M with $c_1(L) = [\omega]$. Choose a connection ∇ on L so that (M, ω, L, ∇) is a prequantum system. Suppose we are given a real polarization of M . Then the quantization of (M, ω, L, ∇) with this real polarization is independent of the choice of ∇ .*

Remark 3.1. Note that in the case where the foliation is given as in [3] by an integrable system, the generic leaves are tori, on which (L, ∇) may have complicated holonomy representations. We have shown that the global condition $H^1(M, \mathbb{R}) = 0$ guarantees these holonomy representations are independent of ∇ .

²Morally $\alpha = t^{-1} dt$, where $t : M \rightarrow S^1$ is a (possibly homotopically nontrivial) element of the gauge group $\text{Map}(M, S^1)$.

Remark 3.2. For complex polarizations, an analog of Corollary 2 holds, since the index of the Dolbeault $\bar{\partial}$ operator on (M, L) depends only on M and $c_1(L) = [\omega]$, not on the connection; this is due to the Riemann-Roch theorem

$$(3.3) \quad \text{ind}(\bar{\partial}) = \int_M Td(TM) e^{[\omega]}.$$

Note that the right-hand side of equation (3.3) depends only on the class $[\omega]$ and not on ∇ .

Remark 3.4. In the case where $H^1(M, \mathbb{R}) \neq 0$, any two connections ∇ and ∇' with the same curvature still give the same quantization if, in addition, the trivial bundle $L \otimes L^*$, equipped with the flat connection arising from ∇ and ∇' , is trivial as a bundle with connection.

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